

TOPIC NAME: Probability

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Joint Probability
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

	alarm	7 alarm
bunglang	0.09	0.01
7 bunglang	0.1	0.8

$$P(\text{alarm} \cap \text{bunglang}) = 0.09$$

$$P(\text{alarm} \cap 7 \text{ bunglang}) = 0.1$$

$$P(7 \text{ bunglang}) = P(7 \text{ bunglang} \cap \text{alarm}) + P(7 \text{ bunglang} \cap 7 \text{ alarm})$$

$$= 0.1 + 0.8 = 0.9$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

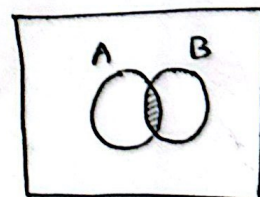
$$P(\text{alarm} | \text{bunglang}) = \frac{P(\text{alarm} \cap \text{bunglang})}{P(\text{bunglang})}$$

$$= \frac{0.09}{0.1}$$

Joint Probability Distribution table: B & A

	B	7B
A	$P(A \cap B)$	$P(A \cap 7B)$
7A	$P(7A \cap B)$	$P(7A \cap 7B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



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	Yes	No
Male	0.32	0.22
Female	0.28	0.18

Given that a person has cheated in the exam, what is the probability that the person is male.

$$\begin{aligned}
 P(\text{male} | \text{yes}) &= \frac{P(\text{male} \cap \text{yes})}{P(\text{yes})} \\
 &= \frac{0.32}{P(\text{yes} \cap \text{male}) + P(\text{yes} \cap \text{female})} \\
 &= \frac{0.32}{0.32 + 0.28} = \frac{0.32}{0.60}
 \end{aligned}$$

Independence

$$P(A \cap B) = P(A) \times P(B)$$

$$(A \perp B)$$



A and B independent

coin toss: $P(H) = 0.5$
 $P(T) = 0.5$

Conditional Independence

$$A \perp B | C$$

studying at home $\perp$ grades in exam | exam (321)

(422)

122 vs 321 exam first (322) grade 0.

so it became independent

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Traffic ↓ late for class | leave home late

$$P(A \cap B | C) = P(A | C) \times P(B | C)$$

* t : toothache

c : cavity

x → x pay

	t		¬t	
	c	¬c	c	¬c
x	0.108	0.012	0.072	0.008
¬x	0.016	0.064	0.144	0.576

a) $P(t \wedge c \wedge x) = 0.108$

b) $P(\neg t \wedge c \wedge \neg x) = 0.144$

c) $P(t \wedge c) = P(t \wedge c \wedge x) + P(t \wedge c \wedge \neg x)$
 $= 0.108 + 0.016$

d) $P(x)$
 $= 0.108 + 0.012 + 0.072 + 0.008$

$$P(A \cap B \cap C) = P(A | C) \times P(B | C) \times P(C)$$

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Bayes Theorem

$$P(A|B)$$

$$1 - P(A|B) = P(\neg A|B)$$

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

* A company manufacture electronic components in 2 factories .

i) Factory A produces 60% of the components and 2% of them are defective

ii) Factory B produces 40% of the components and 5% of them are defective

a) If a randomly selected component is found to be defective, what is the probability that it is produced by factory A.

$$P(\text{factory A} | \text{defective}) = ?$$

Soln: $P(\text{factory A}) = 0.6$

$$P(\text{factory B}) = 0.4$$

$$P(\text{defective} | \text{factory A}) = 0.02$$

$$P(\text{defective} | \text{factory B}) = 0.05$$

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$$P(\text{factory A} | \text{defective}) = \frac{P(\text{defective} | \text{fact A}) \times P(\text{fact A})}{P(\text{defective})}$$

$$= \frac{(0.02) \times (0.6)}{0.032}$$

$$\frac{P(\text{defective} \cap \text{fact A})}{P(\text{defective} \cap \text{fact A}) + P(\text{defective} \cap \text{fact B})}$$

$$= \frac{0.02 \times 0.6}{0.032}$$

$$P(\text{defective} | \text{fact A}) = \frac{P(\text{def} \cap \text{fact A})}{P(\text{fact A})}$$

$$P(\text{defective}) = P(\text{defective} \cap \text{fact A}) + P(\text{def} \cap \text{fact B})$$

$$= P(\text{def} | \text{fact A})P(\text{fact A}) + P(\text{def} | \text{fact B})P(\text{fact B})$$

$$= 0.02 \times 0.6 + 0.05 \times 0.4$$

$$= 0.032$$

* HIV global prevalence = 0.008 $\rightarrow P(\text{HIV}) = 0.008$

$P(T | \text{HIV}) = 95\%$, $P(\neg T | \neg \text{HIV}) = 95\% = 0.95$

Person is likely HIV effected or not?

$P(\text{HIV}) + P(\neg \text{HIV}) = 1 \therefore P(\neg \text{HIV}) = 1 - 0.008 = 0.992$

given $P(T | \text{HIV}) = 0.95$, $P(\neg T | \text{HIV}) = 1 - 0.95 = 0.05$

change $P(\neg T | \neg \text{HIV}) = 0.95$, $P(T | \neg \text{HIV}) = 1 - 0.95 = 0.05$

हलना $P(\text{HIV} | T) = \frac{P(T | \text{HIV}) \cdot P(\text{HIV})}{P(T)} = 0.95 \times 0.008 = 0.0076$

$P(\neg \text{HIV} | T) = \frac{P(T | \neg \text{HIV}) \cdot P(\neg \text{HIV})}{P(T)} = 0.05 \times 0.992 = 0.0496$

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* HIV global prevalence = 0.008

$$P(T|HIV) = 95\%$$

$$P(\neg T|\neg HIV) = 95\%$$

Perform a test the result is positive. What is the probability of having HIV? $P(HIV|T) = ?$

$$P(HIV|T) = \frac{P(T|HIV)P(HIV)}{P(T)}$$

$$\frac{P(T|HIV)}{P(HIV)}$$

$$P(T|HIV)P(HIV)$$

$$P(T|HIV) + P(T|\neg HIV)$$

$$P(T|HIV)P(HIV)$$

$$P(T|HIV)P(HIV) + P(T|\neg HIV)P(\neg HIV)$$

$$0.95 \times 0.008$$

$$= \frac{0.95 \times 0.008}{0.95 \times 0.008 + 0.05 \times 0.992} = 0.1329$$

$$800.0 = 800.0 - 1 = (\text{limiting } r) \text{ } q$$

$$800.0 = (\text{limiting } r) \text{ } q$$

$$500.0 = 800.0 - 1 = (\text{limiting } r) \text{ } q$$

$$\frac{(\text{limiting } r) \text{ } q (\text{limiting } r) \text{ } q}{(\text{limiting } r) \text{ } q} = (\text{limiting } r) \text{ } q$$

$$800.0 \times 800.0 = 800.0 \times 800.0$$

$$\frac{(\text{limiting } r) \text{ } q (\text{limiting } r) \text{ } q}{(\text{limiting } r) \text{ } q} = (\text{limiting } r) \text{ } q$$

$$800.0 \times 800.0 = 800.0 \times 800.0$$

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* Guilty or not ?

A person is put in front of a jury. The jury finds the defendant guilty in 98% of the cases in which the defendant has committed a crime and it finds the defendant not guilty in only 97% of the cases in which the defendant has not committed a crime. Furthermore, only 0.008 of the entire population has committed a crime. If a random person is found guilty by the jury, what's more likely: criminal or not ?

$P(\text{criminal} | \text{guilty})$?

$P(\text{guilty} | \text{criminal}) = 0.98$

$P(\neg \text{guilty} | \text{criminal}) = 1 - 0.98 = 0.02$

$P(\neg \text{guilty} | \neg \text{criminal}) = 0.97$

$P(\text{guilty} | \neg \text{criminal}) = 1 - 0.97 = 0.03$

$P(\text{criminal}) = 0.008$

$P(\neg \text{criminal}) = 1 - 0.008 = 0.992$

$P(\text{criminal} | \text{guilty}) = \frac{P(\text{guilty} | \text{criminal}) P(\text{criminal})}{P(\text{guilty})}$

$= 0.98 \times 0.008 = 7.84 \times 10^{-3}$

$\checkmark P(\neg \text{criminal} | \text{guilty}) = \frac{P(\text{guilty} | \neg \text{criminal}) P(\neg \text{criminal})}{P(\text{guilty})}$

$= 0.03 \times 0.992 = 0.02976$

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Conditional Independence

$$A \perp B \rightarrow C$$

$$P(A \cap B | C) = P(A | C) \times P(B | C)$$

Independence

$$P(A \cap B) = P(A) \times P(B)$$

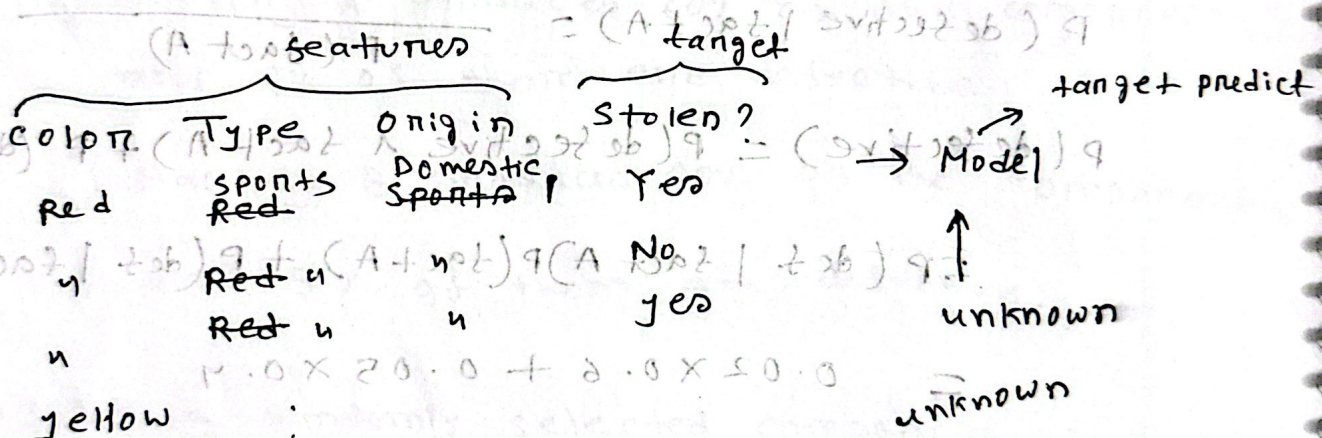
$$A \perp B$$

A → Get chance for higher studies

B → GRE score

C → Supervisor

Naive Bayes Algorithm



color = yellow
Type = sports
Origin = Domestic

Day	Feature				Target
	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Train a Naive Bayes classifier and find out if the player will play tennis.

Given:

Outlook is sunny

Temperature is cool

Humidity is high

Wind speed is strong

TOPIC NAME: _____

* Train a Naive Bayes classifier and find out if the

player will play tennis given:

Outlook is sunny

temperature is cool

humidity is high

and wind speed is strong

Play Tennis

↓
target

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

$$P(\overset{A}{\text{Play}} | \overset{B}{\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong}})$$

$$= \frac{P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong} | \text{Play}) \times P(\text{Play})}{P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong})}$$

$$= \frac{P(\text{sunny} | \text{Play}) \times P(\text{cool} | \text{Play}) \times P(\text{high} | \text{Play}) \times P(\text{strong} | \text{Play}) \times P(\text{Play})}{P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong})}$$

$$= \frac{2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14}{P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong})} = 0.005$$

$$P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong} | \neg \text{Play}) \times P(\neg \text{Play})$$

$$P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong})$$

$$= P(\text{sunny} | \neg \text{Play}) \times P(\text{cool} | \neg \text{Play}) \times P(\text{high} | \neg \text{Play}) \times P(\text{strong} | \neg \text{Play}) \times P(\neg \text{Play})$$

$$= 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.02$$

$$P(A \wedge B | C) = P(A | C) \times P(B | C)$$

Assume there is no dependence $\rightarrow$ but practically not visible (so naive)

$$P(\text{sunny} \wedge \text{cool} \wedge \text{high} \wedge \text{strong} | \text{Play}) = P(\text{sunny} | \text{Play})$$

$$\times P(\text{cool} | \text{Play}) \times P(\text{high} | \text{Play}) \times P(\text{strong} | \text{Play})$$

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Outlook

Humidity

	yes	no
Sunny	$P(\text{Sunny} \text{yes})$ $= 2/9$	$P(\text{Sunny} \text{No})$ $= 3/5$
overcast	$P(\text{OC} \text{yes})$ $= 4/9$	$P(\text{OC} \text{No})$ $= 0/5$
Rain	$P(\text{Rain} \text{yes})$ $= 3/9$	$P(\text{Rain} \text{No})$ $= 2/5$

	yes	no
High	$3/9$	$4/5$
Normal	$6/9$	$1/5$

Temp

Wind

	yes	no
Cool	$3/9$	$1/5$
Mild	$4/9$	$2/5$
Hot	$2/9$	$2/5$

	yes	no
Strong	$3/9$	$3/5$
Weak	$6/9$	$2/5$

$$0.0 = 1/5 \times 2/5 \times 2/5 \times 2/5 \times 2/5 =$$

$$0.0008 \times 0.1 = 0.00008$$